

Programme	B. Sc. STATISTICS				
Course Code	STA 8 EJ 417				
Course Title	ADVANCED TRENDS IN STATISTICS				
Type of Course	Major Elective				
Semester	VIII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Thorough knowledge of probability distributions				
Course Summary	To understand Johnson's system of distributions, Burr family of distributions, Infinite divisibility, U-Statistics & Stochastic ordering.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
1.	Discuss the Johnson's S_B system, Johnson's system S_u and Burr family of distributions.	U	F	Seminar Presentation
2.	Identify the general theory of infinite divisibility and its applications	U	F	Instructor-created exams
3.	Explain the concept and properties of U-statistics, and their role in non-parametric modelling	An	P	Group Tutorial Work
4.	Describe various types stochastic order relations between random variables in univariate setup.	U	C	Home Assignments
* - Remember (R), Understand (U), Apply (Ap), Analyze (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (60)	Marks (70)
I	Distribution Theory		12	15
	1	Systems of distributions		
	2	Johnson's S_B system		
	3	Johnson's S_u system		

	4	Burr distributions		
	5	Infinite divisibility of probability distributions- (i) the non-negative integers		
	6	Infinitely divisible distribution on (ii) the non-negative real's		
II	U-Statistics		12	20
	7	Basic description of U-statistics		
	8	Variance and other moments of a U- statistic		
	9	Projection of a U-statistic on the basic observations		
	10	Almost sure behavior of U-statistics		
	11	Asymptotic distribution theory of U-statistics		
	12	Non-parametric density estimation		
III	Univariate stochastic orders		12	20
	13	Usual stochastic order		
	14	Hazard rate order		
	15	Likelihood ratio order		
	16	Convolution order		
	17	Mean residual life orders		
IV	Univariate variability orders		12	15
	18	Convex order, dispersive order,		
	19	Excess wealth order & peakedness order		
	20	Monotone convex and monotone concave orders		
	21	Transform orders: convex, star orders		
	22	Super additive orders		
V	Open –Ended		12	
	Practical applications of the concepts discussed in Module I to Module IV			

References

1. Laha, R.G. and Rotatgi, V.K. (1979). Probability Theory. Wiley, New York.
2. Serfling, R.J.(1980). Approximation Theorems of Mathematical Statistics (Chapter-5). John Wiley and Sons, Canada.
3. Steutel, F.W. and van Harn, K. (2004). Infinite Divisibility of Probability Distributions on the Real Line. Marcel Dekker Inc., New York.
4. Shaked, M. and Shanthikumar, J. G. (Eds.). (2007). Stochastic Orders. Springer, New York.

MINOR COURSES IN STATISTICS
SYLLABUS